Week 1: Introduction to Optimization

CCS4004 - Optimization Techniques

Amgad A. Salama, Ph.D.

AAST - CCIT

Today's Journey

1. What is Optimization?
2. Types of Optimization Problems
3. Problem Formulation
4. Optimization in Modern Technology
5. Course Roadmap
6. Career Opportunities
7. Let's Practice!
8. Next Week Preview

Quick Question

How many optimization decisions have you made today?

Quick Question

How many optimization decisions have you made today?

- Chose fastest route to university? ✓
- Picked cheapest ride-sharing option? ✓
- Minimized time spent on social media? (Maybe not!)

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What is Optimization?

Definition

Optimization is the process of finding the best solution from all possible solutions.

Key Components:

- **Decision Variables:** What can we control?
- **Objective Function:** What do we want to maximize/minimize?
- **Constraints:** What limitations do we face?

Mathematical Formulation

Maximize or Minimize

$$f(x_1, x_2, \ldots, x_n)$$

$$\downarrow$$

Subject to Constraints

$$g_1(x_1, \ldots, x_n) \leq b_1$$
$$g_2(x_1, \ldots, x_n) \leq b_2$$
$$\vdots$$

Example: Lemonade Stand

Scenario: You want to maximize profit from selling lemonade.

Decision Variables:

- $x =$ cups to make

Objective:

- Maximize profit $= 5x - 2x = 3x$
- (Sell for \$5, cost \$2)

Constraints:

- Sugar: $0.1x \leq 10$ kg
- Lemons: $0.5x \leq 50$
- Time: $0.02x \leq 2$ hours
- $x \geq 0$ (non-negative)

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Question: How many cups should you make?

Real World: Google Search

The Challenge:

- 8.5 billion searches/day
- Rank trillions of web pages
- Must be relevant & fast

Optimization Problem:

- Maximize: User satisfaction
- Subject to: Response time < 0.5s

Search Query

"Best pizza near me"

↓

Optimization

Algorithm

Real World: Amazon Delivery

Problem: Deliver 100 packages across a city.

Optimization Objective

Minimize: Total distance traveled (saves fuel, time, money)

Constraints:

- Each package delivered exactly once
- Truck capacity: 50 packages max
- Delivery time windows (9 AM - 5 PM)
- Traffic conditions

This is called the "Vehicle Routing Problem" (VRP)

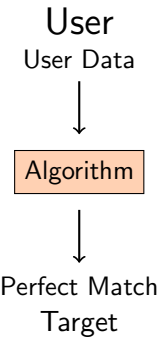
Real World: Netflix Recommendations

The Goal: Keep users watching!

Optimization Problem:

- Maximize: Viewing time per user
- **Variables:** Which shows to recommend
- **Constraints:**
  - User preferences
  - Content availability
  - Diversity of recommendations

Impact: 80% of watched content comes from recommendations!



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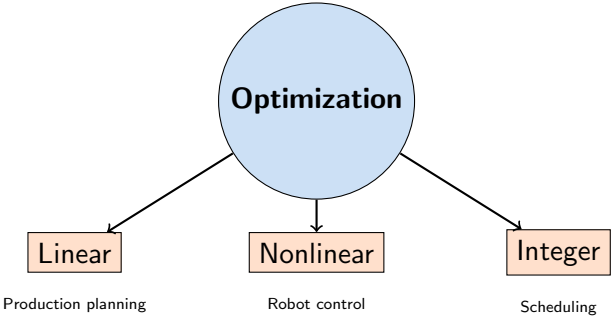
Types of Optimization Problems



Linear Programming (LP)

Characteristics:

- Objective is linear
- Constraints are linear
- Variables are continuous

Example:

$$\begin{aligned} \max \quad & 3x_1 + 5x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 10 \\ & 2x_1 + x_2 \leq 12 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Applications:

- Resource allocation
- Production planning
- Diet optimization
- Portfolio management

Fun Fact

The Simplex method (1947) is one of the top 10 algorithms of the 20th century!

Nonlinear Programming (NLP)

Characteristics:

- Objective or constraints are nonlinear
- More complex than LP
- May have local optima

Example:

$$\begin{aligned} \min \quad & x_1^2 + x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 = 1 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Applications:

- Machine learning training
- Chemical process design
- Power system optimization
- Robotics path planning

Challenge

Finding global optimum is often difficult!

Integer Programming (IP)

Key Feature: Decision variables must be integers!

Example: Binary Decisions

- $x_i = 1$ if project i is selected
- $x_i = 0$ otherwise

Formulation:

$$\begin{aligned} \max \quad & \sum_{i=1}^{n} p_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^{n} c_i x_i \leq B \\ & x_i \in \{0, 1\} \end{aligned}$$

Real Applications:

- Facility location (build or not?)
- Job scheduling (assign or not?)
- Network design (include link or not?)
- Sudoku solving!

Note: Much harder to solve than LP!

Quick Comparison

Aspect	**LP**	**NLP**	**IP**
Linearity	Linear	Nonlinear	Linear/Nonlinear
Variables	Continuous	Continuous	Integer
Difficulty	Easy	Medium	Hard
Solve Time	Fast	Medium	Can be slow
Global Optimum	Guaranteed	Maybe not	Guaranteed*

*If we wait long enough!

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The Art of Formulation

The hardest part isn't solving the problem...

The Art of Formulation

The hardest part isn't solving the problem...

It's formulating it correctly!

Formulation Steps

1. Identify decision variables
2. Define objective function
3. List all constraints
4. Check mathematical correctness

Example: Mobile Data Plan

Problem: A telecom company offers data plans.

Available Plans:

- Plan A: 5 GB for 20 LE
- Plan B: 10 GB for 35 LE
- Plan C: 20 GB for 60 LE

Constraints:

- Company has capacity for 1000 GB total
- At least 30 plans must be sold
- Plan B should be at least 40% of total sales

Goal: Maximize revenue

Mobile Data Plan - Formulation

Step 1: Decision Variables

- x_A = number of Plan A sold
- x_B = number of Plan B sold
- x_C = number of Plan C sold

Step 2: Objective Function

$$\max Z = 20x_A + 35x_B + 60x_C$$

Step 3: Constraints

$$5x_A + 10x_B + 20x_C \leq 1000 \quad \text{(capacity)}$$
$$x_A + x_B + x_C \geq 30 \quad \text{(minimum sales)}$$
$$x_B \geq 0.4(x_A + x_B + x_C) \quad \text{(Plan B requirement)}$$
$$x_A, x_B, x_C \geq 0 \quad \text{(non-negativity)}$$

Continuous vs Discrete Variables

Continuous Variables

- Can take any real value
- Example: Amount of liquid (2.5 liters)
- Easier to solve

Examples:

- Production volume
- Investment amounts
- Temperature settings

Discrete Variables

- Take integer values only
- Example: Number of trucks (can't be 2.5!)
- Harder to solve

Examples:

- Number of employees
- Number of flights
- Yes/No decisions

Choosing the right variable type is crucial!

Common Formulation Mistakes

1. **Forgetting non-negativity constraints**
   - Can't produce -5 units!
2. **Inconsistent units**
   - Mixing kg and pounds, hours and minutes
3. **Missing constraints**
   - Always check: What could go wrong in real life?
4. **Overcomplicating the model**
   - Start simple, add complexity later

Remember: A good model is as simple as possible, but not simpler!

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Optimization Powers Modern Tech

You use optimization every single day!

Your Morning:

- Google Maps route
- Uber pricing
- Email spam filtering
- Spotify playlists

Behind the Scenes:

- Credit card fraud detection
- Factory automation
- Power grid management
- SpaceX launches

Optimization in AI/ML

Training a neural network = Solving an optimization problem!

The Problem

$$\min_{\theta} \quad L(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i)$$

where $\theta =$ network parameters

$L =$ loss function

$f_\theta =$ neural network

Popular Method: Gradient Descent (we'll study this later!)

- ChatGPT: Billions of parameters optimized
- Image recognition: Millions of training examples
- Autonomous driving: Real-time optimization

Case Study: Tesla Production

Challenge: Build cars efficiently at Gigafactory

Optimization Problem:

- Maximize: Cars produced per day
- **Subject to:**
  - Part availability
  - Worker shifts
  - Machine capacity
  - Quality requirements

Results:

- Went from 5,000 to 20,000 cars/week
- Saved millions in costs
- Reduced production time by 40%

Key Insight

Even 1% efficiency gain = $millions saved!

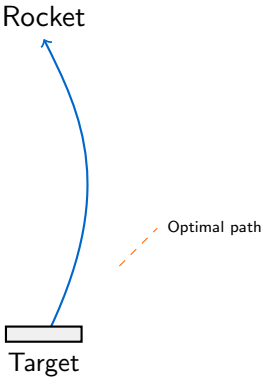
SpaceX: Optimal Trajectories

Problem: Land Falcon 9 rocket on drone ship

Optimization Challenge:

- Minimize: Fuel consumption
- **Subject to:**
  - Physics equations (ODEs!)
  - Safety constraints
  - Landing precision (< 1 meter)
  - Real-time computation

Solution: Nonlinear programming with 100+ variables solved in real-time!



Power Grid Optimization

Problem: Distribute electricity efficiently across a city

Multi-Objective Optimization

- **Minimize:** Generation cost
- **Minimize:** Carbon emissions
- **Maximize:** Grid stability
- **Balance:** Supply = Demand at all times

Challenges:

- Demand changes every second
- Renewable sources are unpredictable (solar, wind)
- Must prevent blackouts
- Thousands of variables

This is a large-scale constrained nonlinear optimization problem!

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What We'll Learn This Semester

Weeks 1-3	Weeks 4-7	Weeks 8-11	Weeks 12-13	Weeks 14-15
Foundations	Linear Programming	Nonlinear Programming	Parsing & Errors	Code Optimization

Our Journey

From basic concepts → Advanced techniques → Real implementations

Course Topics

First Half:

1. ODEs & Optimization
2. Problem Formulation
3. Linear Programming
4. Simplex Method
5. Duality & Sensitivity
6. Multi-objective Optimization
7. Goal Programming

Second Half:

8. Unconstrained NLP
9. Constrained NLP
10. Lagrange Multipliers
11. Barrier Methods
12. Parsing Techniques
13. Error Handling
14. Code Generation
15. Code Optimization

Why Study These Topics?

- **ODEs:** Model dynamic systems (populations, circuits, markets)
- **Linear Programming:** Foundation for many applications
- **Nonlinear Programming:** Real world is rarely linear!
- **Parsing & Code Gen:** Build optimization tools
- **Code Optimization:** Make algorithms run faster

Each topic has direct industry applications!

How We'll Learn

Theory (40%):

- Mathematical foundations
- Algorithm understanding
- Proof intuitions

Practice (60%):

- Real-world examples
- Hands-on exercises
- Coding implementations

Each Week:

- Lecture (theory + examples)
- Tutorial (problem solving)
- Assignment (practice)
- Quiz (mastery check)

Philosophy

"I hear and I forget. I see and I remember. I do and I understand." - Confucius

Tools & Software

Programming Languages:

- Python (primary): NumPy, SciPy, CVXPY
- MATLAB (optional) (It's up to you)

Solvers:

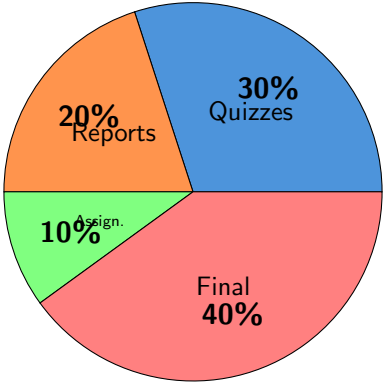
- Open-source: GLPK, COIN-OR
- Commercial: Gurobi (free academic license), CPLEX

Visualization:

- Matplotlib, Plotly
- GeoGebra (for geometric intuition)
- GILP (interactive Simplex visualization)

All tools are free for students!

How You'll Be Evaluated



Quizzes: Weeks 1-7, 8-12, 1-15 — **Reports:** 3 project reports — **Final:** Week 16-17

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Where Will This Take You?

Job Roles Using Optimization:

- **Data Scientist** - Optimize ML models ($90K-150K)
- **Operations Research Analyst** - Supply chain optimization ($80K-130K)
- **Quantitative Analyst** - Financial optimization ($100K-200K+)
- **AI/ML Engineer** - Neural network training ($95K-160K)
- **Software Engineer** - Algorithm optimization ($85K-145K)
- **Industrial Engineer** - Process optimization ($75K-120K)

Optimization skills are highly valued across industries!

Top Companies Using Optimization

Tech Giants:

- Google (search, ads)
- Amazon (logistics)
- Meta (ad targeting)
- Microsoft (cloud resources)
- Netflix (recommendations)

Finance:

- Goldman Sachs
- JPMorgan Chase
- Bloomberg

Transportation:

- Uber (routing, pricing)
- Lyft (matching)
- FedEx (logistics)
- Airlines (scheduling)

Other:

- Tesla (manufacturing)
- SpaceX (trajectory)
- Pharmaceutical companies
- Energy companies

Real Job Postings

Google - Operations Research Analyst

"Design and implement optimization algorithms for resource allocation in data centers. Required: Strong background in linear and nonlinear programming."

Amazon - Supply Chain Optimization Engineer

"Develop optimization models for warehouse operations and delivery routing. Experience with Python and optimization libraries required."

Meta - ML Infrastructure Engineer

"Optimize training pipelines for large-scale neural networks. Understanding of gradient-based optimization essential."

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Pizza Delivery Problem

Scenario: You run a pizza delivery service. You have:

- 5 pizzas to deliver to different locations
- 1 delivery person with a scooter
- Each location is at a different distance
- Customer satisfaction decreases with waiting time

Your Task (3 minutes):

1. What are the decision variables?
2. What should you optimize (objective)?
3. What constraints exist?

Discuss with your neighbor, then we'll share answers!

Pizza Delivery - Solution

Decision Variables:

- Order of deliveries: $\pi = (1, 2, 3, 4, 5)$ (a permutation)
- Or: $x_{ij} = 1$ if we go from location i to j, 0 otherwise

Objective Function:

- Minimize total time: $\min \sum_{i,j} d_{ij} \cdot x_{ij}$
- Or maximize satisfaction: $\max \sum_i s_i(t_i)$ where $s_i =$ satisfaction

Constraints:

- Visit each location exactly once
- Start and end at pizza shop
- Delivery within 30 minutes (cold pizza = unhappy customer!)

This is the famous "Traveling Salesman Problem" (TSP)!

Quick Formulation Exercise

Identify the optimization problem in each scenario:

1. **Student Scheduling:** You have 5 courses to take, each at different times. Minimize breaks between classes.

   Type:

Quick Formulation Exercise

Identify the optimization problem in each scenario:

1. **Student Scheduling:** You have 5 courses to take, each at different times. Minimize breaks between classes.

   Type: Integer Programming (binary: take or not)

Quick Formulation Exercise

Identify the optimization problem in each scenario:

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2. **Diet Planning:** Choose foods to meet nutritional requirements at minimum cost.

   Type:

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   Type: Linear Programming

Quick Formulation Exercise

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1. **Student Scheduling:** You have 5 courses to take, each at different times. Minimize breaks between classes.

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2. **Diet Planning:** Choose foods to meet nutritional requirements at minimum cost.

   Type: Linear Programming

3. **WiFi Router Placement:** Position router to maximize coverage in your house.

   Type:

Quick Formulation Exercise

Identify the optimization problem in each scenario:

1. **Student Scheduling:** You have 5 courses to take, each at different times. Minimize breaks between classes.

   Type: Integer Programming (binary: take or not)

2. **Diet Planning:** Choose foods to meet nutritional requirements at minimum cost.

   Type: Linear Programming

3. **WiFi Router Placement:** Position router to maximize coverage in your house.

   Type: Nonlinear Programming (signal strength is nonlinear!)

When Optimization Goes Wrong!

Famous Optimization Blunders:

Knight Capital (2012)

Trading algorithm error: Lost $440 million in 45 minutes!
Lesson: Test your optimization code thoroughly!

Amazon Pricing Algorithm

Book priced at $23.6 million due to competing pricing algorithms!
Lesson: Always set constraints (upper bounds)!

Airlines Overbooking

Optimized for profit but forgot customer satisfaction constraint!
Lesson: Include all important objectives!

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Next Week: Differential Equations

Week 2 Preview: ODEs in Optimization

What We'll Cover:

- How ODEs model dynamic systems
- Connection between ODEs and optimization
- Numerical methods for solving ODEs
- Applications: Population dynamics, epidemics, circuits

Real Application:

- Model COVID-19 spread
- Optimize vaccination strategies
- Predict epidemic peaks

Prepare

Review: Basic calculus, derivatives, integration

This Week's Tasks

Assignment 1 (Due next week):

1. Formulate 3 optimization problems from daily life
2. For each: identify variables, objective, constraints
3. Classify each as LP, NLP, or IP

Install:

- Python (Anaconda distribution recommended)
- Libraries: numpy, scipy, matplotlib

Learning Resources

Online Resources:

- GILP Tool: `https://gilp.henryrobbins.com`
- YouTube: 3Blue1Brown (calculus visualization)
- Khan Academy: Optimization section

Tips for Success

Do's

- Practice formulation weekly
- Visualize problems geometrically
- Code algorithms yourself
- Form study groups
- Ask questions early
- Connect to real examples

Don'ts

- Don't memorize formulas blindly
- Don't skip fundamentals
- Don't wait until exams
- Don't work in isolation
- Don't ignore code practice
- Don't forget the "why"

Understanding > Memorization

Today's Key Takeaways

1. **Optimization is everywhere** - from Google to your GPS
2. **Three components:** Variables, Objective, Constraints
3. **Three main types:** Linear, Nonlinear, Integer
4. **Formulation is crucial** - Garbage in, garbage out!
5. **Real-world applications** - High-paying jobs await!
6. **This course matters** - Skills you'll use for life

Let's optimize your learning this semester!

"

In theory, theory and practice are the same.
In practice, they are not.

"

- Albert Einstein

That's why we'll do BOTH in this course!

Thank You!

See you next week!

Contact:
Email: [amgad.salama@acm.org]

Start thinking optimization!